

Learning from Itself: Mining Internal Knowledge from Vision Language Models for Continual Learning

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Abstract

Vision-language models like CLIP excel at zero-shot recognition but struggle with continual learning due to two critical issues: (1) severe distribution gap between pretraining captions and post-training class names, and (2) performance mismatch between vision-only and dual-encoder approaches—vision-only methods achieve 20% higher accuracy on fine-grained tasks while CLIP dominates on natural images. We propose *Learning from Itself* (Lfl), which mines CLIP’s internal knowledge to address both challenges. First, we generate pseudo-captions by optimizing learnable tokens to minimize CLIP’s contrastive loss, creating auxiliary training signals that bridge the pretraining-finetuning distribution gap without external models. Second, we introduce adaptive mutual distillation that dynamically weights knowledge transfer between CLIP’s text encoder and a temporary vision classifier based on their instantaneous performance—stronger branches teach more, weaker ones learn more. At inference, only the original CLIP architecture is used, having absorbed discriminative knowledge from both branches. Lfl achieves state-of-the-art results across multiple continual learning benchmarks, demonstrating that CLIP can effectively teach itself to continually learn new tasks. Code is available at: <https://github.com/jordangong/continual-learning>.

1. Introduction

Vision-language models like CLIP [4, 5, 8, 34, 42, 48, 53] have revolutionized how we approach continual learning. By aligning vision and language during pretraining, these models acquire rich semantic knowledge that enables zero-shot recognition—they can classify images they have never seen during training simply by understanding natural language descriptions. Yet when we deploy these models for specialized tasks like medical diagnosis [45] or satellite imagery analysis [21], a fundamental challenge emerges: how do we

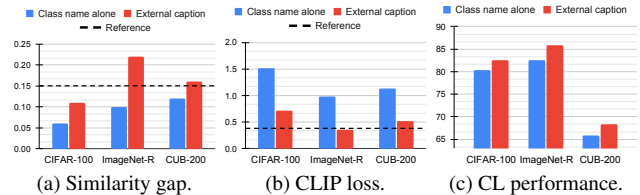


Figure 1. Impact of text distribution on CLIP continual learning (CL). (a) Similarity gap between positive and negative image-text pairs increases substantially when using external descriptive captions versus class names alone. (b) CLIP loss correspondingly decreases, approaching the statistics of natural image-caption datasets (COCO Captions, shown as reference). (c) These improvements translate to 2-3% accuracy gains after continual training across CIFAR-100, ImageNet-R, and CUB-200 datasets.

effectively adapt them to sequential streams of new domains while leveraging their pretrained capabilities?

The standard approaches [47, 54, 57] treat CLIP as just another vision model, discarding its text encoder and applying conventional continual learning techniques. This misses a crucial opportunity. CLIP’s dual-encoder architecture harbors untapped potential—the aligned vision and text encoders together form a knowledge system richer than either component alone [10]. We demonstrate that by mining this internal knowledge, CLIP can essentially teach itself to continually learn new tasks.

Our investigation reveals two critical problems. First, a severe distribution gap exists between pretraining and continual learning in the text space. While CLIP was pre-trained on diverse image-caption pairs scraped from the web [9, 38, 39], continual learning typically provides only class names—reducing rich descriptions to stark labels like “dog” or “car.” We quantify this gap by measuring two key metrics on the pretrained CLIP before any adaptation (Fig. 1). The similarity gap measures the difference between positive (matched) and negative (unmatched) image-text pairs—a larger gap indicates better alignment between modalities (Fig. 1a). The CLIP loss (contrastive loss) measures how well the pretrained model aligns with the target data—lower values suggest less distribution shift (Fig. 1b). On CIFAR-

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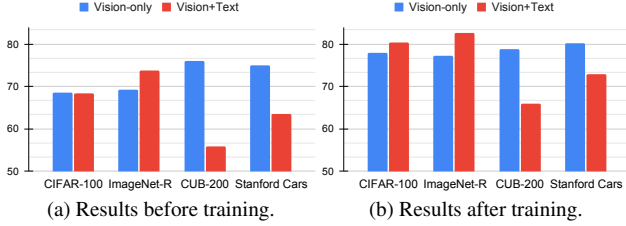


Figure 2. Performance comparison between vision-only and full CLIP approaches (Vision+Text). Vision-only methods excel on fine-grained datasets (CUB-200 and Stanford Cars) both before and after training, while CLIP dominates on natural image datasets (CIFAR-100 and ImageNet-R).

100 with class names only, we observe a similarity gap of 0.06 and CLIP loss of 1.51, far from the reference COCO Captions dataset [3] (gap: 0.15, loss: 0.38). This mismatch signals severe distribution shift. When we replace class names with descriptive captions generated by multimodal language models [50], the gap nearly doubles (0.06→0.11) and the loss halves (1.51→0.71), approaching the reference statistics and indicating reduced distribution shift. After continual learning, this alignment translates directly to performance: descriptive captions improve accuracy by 2-3% across multiple benchmarks (Fig. 1c). But relying on external models for caption generation violates the fundamental principle—the model should learn from its own capabilities, not depend on more powerful systems.

Second, we discover an unexpected phenomenon: sometimes CLIP’s own vision encoder, when used independently, outperforms the full model (Fig. 2). To investigate this, we extract a prototypical classifier from CLIP’s vision encoder itself using SimpleCIL [57]—a training-free method that constructs class prototypes by averaging the model’s own feature representations. This vision-only approach, derived entirely from CLIP’s internal representations, achieves remarkable zero-shot performance (Fig. 2a): on fine-grained datasets like CUB-200, it reaches 76% accuracy compared to full CLIP’s 56%—a striking 20% gap. This reveals that CLIP already contains complementary knowledge within its different components: the vision encoder excels at fine-grained visual distinctions while the text encoder captures broader semantic relationships. After training, this pattern persists (Fig. 2b): vision-only methods excel on fine-grained tasks while CLIP dominates on natural image datasets. Simply ensembling both predictions would work, but it breaks its elegant architecture and prevents downstream integration.

Our solution, Learning from Itself (Lfi), enables CLIP to mine and reorganize its own internal knowledge. For the text distribution gap, we propose a counterintuitive approach: let CLIP generate its own “captions.” We optimize learnable tokens for each image by minimizing CLIP loss while the model remains frozen, essentially asking CLIP to describe what it sees using its pretrained knowledge. These tokens, generated entirely from CLIP’s internal understanding and

regularized to stay within the vocabulary distribution, serve as pseudo-captions that bridge pretraining-finetuning gap.

For the vision-text performance mismatch, we enable CLIP’s components to teach each other through adaptive mutual distillation. During training, we extract a temporary classifier from the vision encoder’s own features—using its pretrained representations to create class prototypes. This creates two prediction heads, both derived from CLIP itself: the original dual-encoder and the prototypical classifier from its vision branch. Rather than treating them equally, we let them learn from each other adaptively—the better-performing component teaches more, allowing CLIP to internally transfer knowledge between its own specialized parts. At inference, we discard the auxiliary classifier and use only the original CLIP architecture, having successfully mined and integrated knowledge from within itself.

We validate Lfi through extensive experiments across diverse continual learning benchmarks, testing on both natural image datasets [6, 13, 23] and fine-grained classification tasks [1, 22, 43]. Our method consistently outperforms existing approaches, achieving new state-of-the-art results.

Our contributions are threefold:

- We identify and quantify the text distribution gap in CLIP continual learning, then mitigate it by mining pseudo-captions from CLIP’s own knowledge.
- We reveal the vision-text performance mismatch and propose adaptive mutual distillation to combine both edges.
- We achieve state-of-the-art results across diverse continual learning benchmarks, from natural to fine-grained images.

2. Related Work

Continual learning. Continual learning addresses the challenge of learning sequential tasks without catastrophic forgetting [32]. Traditional methods training from scratch include regularization-based approaches that constrain parameter updates through distillation or importance weighting [20, 24, 29, 30], architecture-based methods that dynamically expand networks [37, 51], and replay-based techniques that store exemplars [2, 26, 35]. The emergence of pretrained models has shifted focus to adaptation strategies. Parameter-efficient tuning methods include prompt-based approaches [40, 46, 47] that learn task-specific instructions, general PEFT frameworks [11] compatible with Adapter/LoRA/Prefix tuning, interference-free LoRA subspaces [25], and training-free random projections [33]. Representation-based methods leverage pretrained features directly: SLCA [54] addresses progressive overfitting through selective learning rate reduction, while SimpleCIL and APER [57] demonstrate that prototypical classifiers from pretrained features already achieve strong zero-shot performance.

Continual learning with vision language models. Vision-language models enable continual learning through natural language descriptions, but adapting them requires preserving

both pretrained knowledge and vision-language alignment. Early works adapt prompting to CLIP’s multimodal setting, with S-Prompts [44] learning domain-specific prompts independently. Preserving zero-shot capabilities is critical: ZSCL [56] uses distillation with reference data, MoE-Adapter [52] employs mixture-of-experts with automatic routing, while continual pretraining studies [7, 28, 36, 49] explore temporal model merging and domain adaptation strategies. Various architectural modifications leverage CLIP’s dual-encoder structure: MagMax [31] uses only vision encoder with frozen text embeddings, PROOF [59] trains task-specific projections for both modalities, and RAPF [15] performs adaptive representation adjustment with parameter fusion. Recent insights reveal important phenomena: MG-CLIP [16] identifies modality gap as a knowledge preservation indicator, DMNSP [18] discovers asymmetric parameter distributions motivating modality-specific treatment, and Continual-CLIP [41] shows frozen CLIP already achieves strong zero-shot performance. Advanced methods like ENGINE [58] inject external knowledge from GPT-4, while CLAP [17] introduces probabilistic modeling for uncertainty estimation. Unlike the approaches requiring external models, our method mines CLIP’s internal knowledge through self-generated training signals and mutual teaching between encoders.

3. Preliminary

Continual learning problem. Continual learning aims to sequentially learn from a stream of tasks without forgetting previously acquired knowledge. Formally, we consider a sequence of T tasks $\{\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_T\}$, where each task $\mathcal{T}_t$ consists of a dataset $\mathcal{D}_t = \{(x_i^t, y_i^t)\}_{i=1}^{N_t}$ with N_t samples. Here, $x_i^t \in \mathcal{X}$ represents an input image and $y_i^t \in \mathcal{Y}_t$ denotes its corresponding label from the task-specific label set $\mathcal{Y}_t$. The challenge lies in the constraint that when learning task $\mathcal{T}_t$, the model f only has access to the current task’s data $\mathcal{D}_t$, while previous tasks’ data are unavailable. The objective is to minimize the expected loss across all seen tasks. In class-incremental learning, which we focus on, the label spaces are disjoint: $\mathcal{Y}_i \cap \mathcal{Y}_j = \emptyset$ for $i \neq j$, and the model must classify among all classes seen so far: $\mathcal{Y}_{1:t} = \bigcup_{k=1}^t \mathcal{Y}_k$.

For vision-language models like CLIP, we extend this formulation to dual-encoder architectures. Let $f_v : \mathcal{X} \rightarrow \mathbb{R}^d$ be the vision encoder mapping images to d -dimensional embeddings, and $f_l : \mathcal{S} \rightarrow \mathbb{R}^d$ be the text encoder mapping text sequences from space $\mathcal{S}$ to the same embedding space. The encoders produce L2-normalized embeddings: $z^v = f_v(x)/\|f_v(x)\|_2$ and $z^l = f_l(s)/\|f_l(s)\|_2$. Given an image x and a set of textual class descriptions $\{s_c\}_{c \in \mathcal{Y}_{1:t}}$ where $s_c \in \mathcal{S}$ is the text description for class c , CLIP performs zero-shot classification:

$$p(y = c|x) = \frac{\exp(z^v \cdot z_c^l / \tau)}{\sum_{c' \in \mathcal{Y}_{1:t}} \exp(z^v \cdot z_{c'}^l / \tau)}, \quad (1)$$

where $\tau > 0$ is a temperature parameter.

Prototypical classifier. It offers a training-free approach to continual learning by representing each class as a prototype—a representative point in the feature space. Following SimpleCIL [57], we extract class prototypes using the averaged normalized features from the pretrained vision encoder. For class c in task $\mathcal{T}_t$, the prototype is computed as:

$$\mu_c = \frac{1}{|\mathcal{D}_t^c|} \sum_{(x,y) \in \mathcal{D}_t^c} z^v, \quad (2)$$

where $\mathcal{D}_t^c = \{(x, y) \in \mathcal{D}_t : y = c\}$ represents all samples of class c in the current task, and $z^v = f_v(x)/\|f_v(x)\|_2$ is the normalized vision embedding.

In our method, we leverage this prototypical initialization differently. At the beginning of each task $\mathcal{T}_t$, we create a task-specific cosine classifier $g_t : \mathbb{R}^d \rightarrow \mathbb{R}^{|\mathcal{Y}_t|}$ with weight matrix $W_t \in \mathbb{R}^{|\mathcal{Y}_t| \times d}$, where each row w_c is initialized as $w_c = \mu_c$ for $c \in \mathcal{Y}_t$. This classifier is trained only on the current task’s classes and discarded after task completion. Unlike traditional continual learning approaches that maintain a growing classifier, our task-specific prototypical classifier serves purely as a training-time auxiliary that helps distill visual discriminative knowledge into the CLIP model.

Mutual learning. It enables multiple networks to learn collaboratively by treating each network as both student and teacher [55]. In the classical formulation with two models producing probability distributions p_1 and p_2 over the same label space, mutual learning minimizes:

$$\mathcal{L}_{\text{mutual}} = D_{\text{KL}}(p_1||p_2) + D_{\text{KL}}(p_2||p_1), \quad (3)$$

where $D_{\text{KL}}(p||q) = \sum_i p_i \log(p_i/q_i)$ is the Kullback-Leibler (KL) divergence measuring how p differs from q .

This symmetric formulation assumes equal model reliability—valid when training from scratch but problematic with pretrained models where performance asymmetries exist. CLIP’s vision encoder excels at fine-grained visual distinctions while the text encoder captures semantic relationships, with relative strengths varying by dataset. Symmetric mutual learning would force the stronger model to match weaker predictions, degrading performance, necessitating an asymmetric formulation that allows the stronger model to guide while preserving its own predictions.

4. Method

4.1. Pipeline

Learning from Itself enhances CLIP’s continual learning by mining knowledge from the model itself. In Fig. 3, our training pipeline simultaneously optimizes three complementary objectives that address the challenges in continual learning. **Adaptive Mutual Training (Fig. 3a).** When learning a new task, we maintain two classification heads: the standard

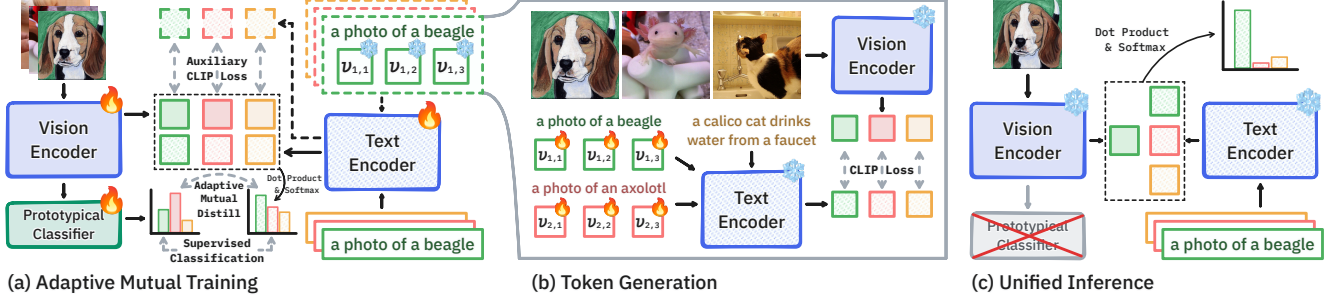


Figure 3. Learning from Itself training pipeline. (a) Adaptive Mutual Training: Our core training simultaneously optimizes three objectives. Supervised classification with ground truth labels (class names) provides discriminative power for both CLIP and a prototypical vision classifier. Auxiliary CLIP loss uses generated tokens (see zoomed detail) to smooth distribution shift. Adaptive mutual distillation between classification heads transfers complementary knowledge, weighted by instantaneous performance. (b) Token Generation (zoomed detail): Before each task, we freeze the original pretrained CLIP and optimize sample-specific tokens $\{v_{i,1}, v_{i,2}, v_{i,3}\}$ as pseudo-captions. Target data is mixed with COCO Captions for contrastive richness. These tokens are then used in the auxiliary CLIP loss during training. (c) Unified Inference: The prototypical classifier is discarded, leaving only the enhanced CLIP model for standard inference with class name templates. Fire and snow symbols indicate trainable and frozen parameters.

CLIP head that classifies via vision-language alignment, and a prototypical vision classifier initialized from the pretrained vision encoder. These heads learn through three parallel objectives: (1) *Supervised Classification*: Both heads are trained with ground truth labels using class names, providing the discriminative signals necessary for classification. (2) *Auxiliary CLIP Loss*: We use sample-specific generated tokens (detailed below) as pseudo-captions in an auxiliary CLIP loss, recreating a pretraining-like environment that smooths the distribution shift from web captions to classification labels. (3) *Adaptive Mutual Distillation*: The two classification heads teach each other, with the teaching strength dynamically weighted by their instantaneous performance, allowing the better-performing head to guide the weaker one.

The key insight is that these objectives are complementary. Supervised classification provides discrimination but suffers from distribution mismatch with CLIP’s pretraining. The auxiliary CLIP loss with generated tokens bridges this distribution gap but lacks explicit class boundaries. Mutual distillation transfers specialized knowledge between the vision and language branches. Together, they enable stable and effective adaptation.

Token generation (Fig. 3b). Before training on each task, we generate sample-specific tokens that serve as pseudo-captions. We freeze the original pretrained CLIP and optimize only learnable tokens to minimize contrastive loss, effectively mining CLIP’s internal knowledge about what textual description best matches each image. This process is task-independent—the same pretrained CLIP is used regardless of task order. We mix target data with COCO Captions during generation to ensure rich contrastive learning, and apply vocabulary alignment and supervised contrastive regularizations to prevent adversarial solutions. These generated tokens are then used in the auxiliary CLIP loss during actual training, providing the pretraining-like signal that comple-

ments classification training.

Unified inference (Fig. 3c). After training, we discard the prototypical classifier and predict using only the standard CLIP architecture with class name templates. The enhanced CLIP model has absorbed discriminative knowledge through mutual distillation while maintaining vision-language alignment through auxiliary contrastive learning, providing improved classification without architectural modifications.

4.2. CLIP as Image “Captioner”

The text distribution gap between pretraining and continual learning severely limits CLIP’s adaptation capability—CLIP sees rich, descriptive captions during pretraining (“a golden retriever playing fetch in the park on a sunny day”) but only class names during continual learning (“dog”), damaging the nuanced vision-language alignment. We propose letting CLIP generate its own captions by optimizing in its continuous token embedding space to find text representations that maximize alignment with given images, analogous to adversarial examples but instead finding text embeddings that best describe images according to CLIP’s own understanding.

For each image x_i in the current task $\mathcal{T}_t$, we create K learnable token embeddings $\mathbf{V}_i = \{v_{i,1}, v_{i,2}, \dots, v_{i,K}\}$ where $v_{i,k} \in \mathbb{R}^d$. These tokens are inserted into a template that preserves class information:

$$s_i = \text{a photo of a } [\text{name}_c] [v_{i,1}] [v_{i,2}] \dots [v_{i,K}], \quad (4)$$

where $[\text{name}_c]$ represents the tokenized class name for class $c = y_i$, providing semantic anchoring.

Initialization strategy. Random initialization would require extensive optimization and might converge to adversarial solutions. Instead, we initialize each $v_{i,k}$ by cycling through the class name’s token embeddings:

$$v_{i,k}^{(0)} = \text{embed}(\text{name}_c)[k \bmod |\text{name}_c|], \quad (5)$$

where $\text{embed}(\cdot)$ maps text to token embeddings. This grounds the optimization in relevant semantic space.

Contrastive optimization. Given the limited class diversity in each continual learning step, we augment each batch with samples from COCO Captions dataset $\mathcal{D}_{\text{ref}}$ to provide rich negative samples. For a mixed batch $\mathcal{B} = \mathcal{B}_t \cup \mathcal{B}_{\text{ref}}$ where $\mathcal{B}_t \subset \mathcal{D}_t$ and $\mathcal{B}_{\text{ref}} \subset \mathcal{D}_{\text{ref}}$, we minimize:

$$\mathcal{L}_{\text{CLIP}} = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \log \frac{\exp(z_i^v \cdot z_i^l / \tau)}{\sum_{j \in \mathcal{B}} \exp(z_i^v \cdot z_j^l / \tau)}, \quad (6)$$

where $z_i^v = f_v(x_i) / \|f_v(x_i)\|_2$ and $z_i^l = f_l(s_i) / \|f_l(s_i)\|_2$ are normalized embeddings.

Regularization for semantic coherence. Pure contrastive optimization often produces adversarial tokens—meaningless patterns that minimize loss but destroy semantic structure [12]. We add two crucial regularizations: (1) *Supervised Contrastive Loss* [19] ensures same-class samples maintain similar text representations:

$$\mathcal{L}_{\text{SCL}} = -\frac{1}{|\mathcal{B}_t|} \sum_{i \in \mathcal{B}_t} \frac{1}{|\mathcal{P}_i|} \sum_{p \in \mathcal{P}_i} \log \frac{\exp(z_i^l \cdot z_p^l / \tau)}{\sum_{k \in \mathcal{B}_t \setminus \{i\}} \exp(z_i^l \cdot z_k^l / \tau)}, \quad (7)$$

where $\mathcal{P}_i = \{j : y_j = y_i, j \neq i\}$ contains same-class samples. (2) *Vocabulary alignment loss* constrains tokens to remain within the natural language distribution. Let $\mathcal{V}$ denote the vocabulary of all token embeddings in the text encoder. For each token, we find its M nearest vocabulary neighbors and minimize their distance:

$$\mathcal{L}_{\text{vocab}} = \frac{1}{|\mathcal{B}_t| \cdot K} \sum_{i \in \mathcal{B}_t} \sum_{k=1}^K \min_{w \in \text{top-}M(\mathcal{V}, v_{i,k})} (1 - \cos(v_{i,k}, w)), \quad (8)$$

This objective prevents drift into adversarial space. Finally, the complete token generation objective combines all three losses: $\mathcal{L}_{\text{gen}} = \mathcal{L}_{\text{CLIP}} + \lambda_{\text{SCL}} \mathcal{L}_{\text{SCL}} + \lambda_{\text{vocab}} \mathcal{L}_{\text{vocab}}$, where λ_{SCL} and λ_{vocab} are hyperparameters. We optimize only $\{\mathbf{V}_i\}$ while keeping all CLIP parameters frozen. These tokens then serve as inputs for the auxiliary CLIP loss $\mathcal{L}_{\text{CLIP}}^{\text{aux}}$ during the actual training phase, helping to bridge the distribution gap between pretraining and finetuning while the primary classification loss provides discriminative signals.

4.3. Mutual Learning between Vision and Language

The vision encoder and text encoder in CLIP possess complementary knowledge. The vision encoder captures fine-grained visual patterns crucial for distinguishing similar-looking objects, while the text encoder understands semantic relationships and conceptual hierarchies. Our goal is to combine these strengths without architectural changes.

Dual-head architecture. During training of task $\mathcal{T}_t$, we maintain two prediction heads that both operate on the current task’s classes: *CLIP head* uses both encoders to classify

via vision-language alignment for current task classes:

$$p_{\text{clip}}(c|x) = \frac{\exp(z^v \cdot z_c^l / \tau)}{\sum_{c' \in \mathcal{Y}_t} \exp(z^v \cdot z_{c'}^l / \tau)}, \quad (9)$$

where z_c^l is the normalized text embedding for class c using template “a photo of a [class name]”. While *vision head* uses only the vision encoder with a task-specific cosine classifier:

$$p_{\text{vis}}(c|x) = \text{softmax}(g_t(f_v(x)) / \tau)_c, \quad (10)$$

where g_t is the prototypical classifier freshly initialized for task $\mathcal{T}_t$ using prototypes $\{\mu_c\}_{c \in \mathcal{Y}_t}$ extracted from the original pretrained vision encoder before continual learning.

Adaptive distillation mechanism. Traditional mutual learning employs bidirectional KL divergence treating both predictions equally, which fails with pretrained models where one consistently outperforms the other. While unidirectional distillation from the stronger to weaker head seems intuitive, it faces critical limitations: we cannot know a priori which head performs better (this varies dramatically across datasets and even samples), and it ignores valuable knowledge in the “weaker” head that may excel on certain samples or capture complementary aspects.

Our solution is to create a single adaptive target that dynamically combines both predictions based on their instantaneous quality. For each sample, we first compute cross-entropy losses to measure prediction quality: $\ell_{\text{clip}} = \text{CE}(p_{\text{clip}}, y)$ and $\ell_{\text{vis}} = \text{CE}(p_{\text{vis}}, y)$, where $\text{CE}(\cdot, \cdot)$ denotes cross-entropy loss and y is the ground truth label. These losses directly measure prediction quality—lower loss means better prediction. We then construct an adaptive distillation target. Critically, we apply stop-gradient to both predictions before mixing to prevent the gradient flow from creating trivial solutions where both heads converge to the same point:

$$q = \alpha \cdot \text{sg}(p_{\text{clip}}) + (1 - \alpha) \cdot \text{sg}(p_{\text{vis}}), \quad (11)$$

The mixing coefficient α is determined by the relative performance difference:

$$\alpha = \sigma \left(\gamma \cdot \frac{\ell_{\text{vis}} - \ell_{\text{clip}}}{\ell_{\text{vis}} + \ell_{\text{clip}} + \epsilon} \right). \quad (12)$$

Here σ is the sigmoid function, γ controls the sharpness of the transition, and ϵ prevents division by zero. When $\ell_{\text{clip}} < \ell_{\text{vis}}$ (CLIP head performs better), $\alpha > 0.5$ and the shared distillation target is dominated by CLIP’s prediction; conversely, when vision performs better, $(1 - \alpha) > 0.5$. This creates an elegant self-balancing system: both heads learn toward a target dominated by the better performer.

Bidirectional knowledge transfer. The mutual distillation losses guide both predictions toward the same adaptive target: $\mathcal{L}_{\text{dist}}^{\text{vis}} = D_{\text{KL}}(p_{\text{vis}} \| q)$, and $\mathcal{L}_{\text{dist}}^{\text{clip}} = D_{\text{KL}}(p_{\text{clip}} \| q)$. This formulation ensures that the weaker head learns more (larger KL divergence when its prediction differs from the target) while the stronger head is gently refined (smaller KL divergence as the target is already close to its prediction).

Table 1. Continual learning performance on OpenCLIP ViT-B/16. Results across five datasets with 10-task and 6-task protocols.

Method	CIFAR-100				ImageNet-R		CUB-200				Stanford Cars		Food-101	
	10-10 (10 tasks)		50-10 (6 tasks)		20-20 (10 tasks)		20-20 (10 tasks)		100-20 (6 tasks)		20-20 (10 tasks)		20-20 (10 tasks)	
	<i>last</i>	<i>avg</i>	<i>last</i>	<i>avg</i>	<i>last</i>	<i>avg</i>	<i>last</i>	<i>avg</i>	<i>last</i>	<i>avg</i>	<i>last</i>	<i>avg</i>	<i>last</i>	<i>avg</i>
L2P [47]	73.03	82.74	73.61	81.14	66.52	75.97	57.93	70.87	66.12	75.64	61.82	76.63	77.33	85.66
DualPrompt [46]	72.44	81.63	72.57	80.12	66.65	76.21	57.46	69.89	64.84	74.40	62.94	76.26	77.29	84.92
CODA-P [40]	73.43	82.43	71.58	78.69	68.95	77.69	62.98	73.12	62.21	73.95	66.47	80.21	78.78	86.18
SimpleCIL [57]	76.68	81.41	76.68	78.29	74.05	79.18	79.57	86.17	79.57	82.42	86.95	91.01	85.73	90.04
Cont-CLIP [41]	71.11	77.86	71.11	73.57	74.53	79.22	64.15	76.70	64.15	69.94	79.69	84.84	85.42	89.75
Finetune	80.91	87.33	83.01	86.40	81.29	85.57	71.06	82.73	73.14	79.48	84.49	89.73	87.48	91.66
CoOp [60]	69.83	78.77	73.15	77.62	75.96	81.38	62.32	75.33	66.28	73.18	80.23	86.26	85.21	90.24
LoRA [14]	79.69	86.45	81.96	85.39	80.58	85.53	69.52	81.41	72.33	78.79	84.14	89.43	87.00	91.39
RAPF [15]	78.04	86.14	77.93	82.17	70.48	81.26	62.77	79.09	62.93	72.82	62.85	82.89	81.15	88.57
PROOF [59]	79.05	86.70	78.87	82.92	80.10	85.34	79.43	84.93	79.18	81.67	89.84	93.26	84.73	90.04
ENGINE [58]	79.22	86.92	79.47	83.15	80.37	86.22	80.20	86.65	79.30	82.59	90.08	94.14	83.89	89.81
Lfl (ours)	83.82	88.65	85.57	87.48	83.57	86.25	81.74	87.23	82.07	86.18	91.42	94.41	89.44	92.47
<i>Joint</i>	89.44	—	89.44	—	86.07	—	83.64	—	83.64	—	92.69	—	91.87	—

4.4. Complete Training Objective

The final training combines supervised learning, auxiliary contrastive learning with generated tokens, and adaptive mutual distillation:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{CE}}^{\text{clip}} + \mathcal{L}_{\text{CE}}^{\text{vis}}}_{\text{Supervised}} + \underbrace{\beta_1 \mathcal{L}_{\text{CLIP}}^{\text{aux}}}_{\text{Auxiliary Contrastive}} + \underbrace{\beta_2 (\mathcal{L}_{\text{dist}}^{\text{vis}} + \mathcal{L}_{\text{dist}}^{\text{clip}})}_{\text{Mutual Distillation}}, \quad (13)$$

where $\mathcal{L}_{\text{CE}}^{\text{clip}} = \text{CE}(p_{\text{clip}}, y)$ and $\mathcal{L}_{\text{CE}}^{\text{vis}} = \text{CE}(p_{\text{vis}}, y)$ are cross-entropy losses with ground truth labels providing the primary discriminative signal, $\mathcal{L}_{\text{CLIP}}^{\text{aux}}$ is the auxiliary CLIP loss using our generated tokens to smooth the pretraining-finetuning distribution gap, and β_1, β_2 are hyperparameters.

After task $\mathcal{T}_t$ completes, we discard the task-specific vision classifier g_t . For inference, the enhanced CLIP model uses standard text templates (“a photo of a [class name]”) rather than generated tokens, computing probabilities across all seen classes as in Eq. (1). This avoids train-test mismatch—generated tokens serve as auxiliary training signals to bridge distribution gaps, while inference maintains standard CLIP deployment protocol.

5. Experiments

5.1. Setup

Datasets. We evaluate on six benchmark datasets across two categories: natural image classification (CIFAR-100 [23] with 100 classes of 32×32 low-resolution images, ImageNet-R [13] with 200 classes exhibiting domain shift through artistic renditions, and ImageNet-100, a 100-class subset of ImageNet-1K [6] and fine-grained visual recognition (CUB-200 [43] with 200 bird species, Stanford Cars [22] with 196 car models, and Food-101 [1] with 101 food categories). Following standard protocols [16, 58, 59], we partition datasets using B - I notation where B denotes base classes and I incremental classes: uniform splitting (*e.g.*, 10-10 for 10-task

CIFAR-100) and base-incremental setup with 50% initial classes followed by 5 equal steps (*e.g.*, 50-10 for 6 tasks).

Baseline methods. We compare against state-of-the-art methods across multiple categories on two CLIP variants (OpenAI CLIP ViT-B/16 [34] and OpenCLIP ViT-B/16 trained on LAION-400M [4, 38]): vision-only adaptations including prompt-based methods (L2P [47], DualPrompt [46], CODA-Prompt [40]), prototypical approaches (SimpleCIL, APER [57]), and SLCA [54] with selective learning rates; vision-language methods including Continual-CLIP [41] (zero-shot baseline), CoOp [60], LoRA [14], MagMax [31], CLAP [17], MG-CLIP [16], PROOF [59], RAPF [15], and ENGINE [58]; plus naive finetuning as reference.

Evaluation metrics. We report final accuracy (*last*) measuring classification performance on all seen classes after the final task, and average accuracy (*avg*) computed as mean accuracy across all continual learning steps, reflecting the stability-plasticity balance throughout the learning process.

Implementation details. For token generation, we freeze CLIP and optimize $K = 3$ learnable tokens per sample using AdamW [27] (learning rate 10^{-3} , weight decay 0.2, batch size 1024 with 512 target + 512 COCO reference samples) for 5 epochs with early stopping, setting $\lambda_{\text{SCL}} = \lambda_{\text{vocab}} = 1.0$ and $M = 5$ vocabulary neighbors. During continual training, we unfreeze all CLIP parameters except temperature ($\tau \approx 0.01$), training with SGD (learning rate 10^{-4} , momentum 0.9, batch size 64) for 20 epochs, setting mixing sharpness $\gamma = 3$, auxiliary CLIP weight $\beta_1 = 0.1$, and mutual distillation weight $\beta_2 = 1.0$, using standard augmentations (random resized crop, horizontal flip, ImageNet normalization).

5.2. Benchmark Results

Tab. 1 and Tab. 2 present comprehensive continual learning results on OpenCLIP and OpenAI CLIP ViT-B/16 backbones, with Lfl consistently outperforming other methods.

Table 2. Continual learning performance on OpenAI CLIP ViT-B/16. Results on three natural image datasets with 10-task protocol.

Method	CIFAR-100		ImageNet-R		ImageNet-100	
	last	avg	last	avg	last	avg
L2P [47]	73.08	81.90	75.98	81.67	67.22	80.51
DualPrompt [46]	72.51	81.45	75.77	82.01	67.38	80.65
CODA-P [40]	62.25	76.98	67.52	78.00	34.76	64.13
SLCA [54]	67.58	80.53	70.37	75.92	59.92	78.63
APER [57]	65.50	75.76	71.35	78.65	76.40	85.84
MagMax [31]	79.00	85.63	80.85	87.13	75.92	86.33
SimpleCIL [57]	68.51	74.36	69.20	76.04	75.46	84.96
Cont-CLIP [41]	68.36	74.64	73.73	79.17	72.64	83.79
Finetune	80.31	86.00	82.56	87.34	77.68	87.21
CoOp [60]	69.84	78.14	78.12	83.34	73.36	84.58
LoRA [14]	79.18	85.48	82.19	87.11	77.02	86.69
RAPF [15]	79.04	86.19	80.28	85.58	80.23	87.51
CLAP [17]	78.21	86.13	79.98	85.77	79.16	87.76
PROOF [59]	76.29	84.88	77.05	82.83	72.48	84.71
MG-CLIP [16]	80.57	87.00	82.67	87.58	78.38	87.31
Lfl (ours)	82.69	87.84	83.97	88.08	80.68	88.29
Joint	89.02	—	86.45	—	86.74	—

As shown in Tab. 1, our method achieves state-of-the-art performance across all datasets and protocols, with particularly strong improvements on fine-grained tasks—reaching 91.42% on Stanford Cars (surpassing ENGINE by 1.34%) and 81.74% on CUB-200 (1.54% over ENGINE). On natural image datasets, Lfl maintains consistent superiority with 83.82% on CIFAR-100 and 83.57% on ImageNet-R, approaching joint training performance. Surprisingly, naive finetuning demonstrates competitive results (80.91% on CIFAR-100) when properly configured with fixed temperature and current-task-only text templates. This suggests that CLIP’s dual-encoder architecture requires different adaptation strategies than vision-only models.

Cross-model evaluation on OpenAI CLIP confirms the robustness of our approach (Tab. 2), with Lfl achieving 82.69% on CIFAR-100, 83.97% on ImageNet-R, and 80.68% on ImageNet-100, consistently outperforming all baselines including recent vision-language methods MG-CLIP. The performance patterns mirror those observed with OpenCLIP—finetuning remains surprisingly effective (82.56% on ImageNet-R), while prompt-based methods struggle, with CODA-Prompt catastrophically failing on ImageNet-100 (34.76%). Vision-only adaptations like SimpleCIL show strong zero-shot capabilities but limited improvement potential, while methods leveraging both encoders achieve better final performance but still fall short of our approach by 1-3%.

5.3. Ablation Studies

All ablations are conducted using OpenCLIP ViT-B/16 on CIFAR-100 (natural images) and CUB-200 (fine-grained classification) with 10-task protocols.

Impact of individual learning components. Tab. 3 dis-

Table 3. Ablation of learning objectives. Performance with combinations of supervised, auxiliary contrastive, and distillation losses.

Loss	CIFAR-100		CUB-200	
	last	avg	last	avg
N/A (a.k.a Zero-shot)	71.11	77.86	64.15	76.70
w/ $\mathcal{L}_{CE}^{clip}$ (a.k.a Finetune)	80.91	87.33	71.06	82.73
w/ $\mathcal{L}_{CLIP}^{aux}$	75.83	80.85	66.29	78.40
w/ $\mathcal{L}_{CE}^{clip} + \mathcal{L}_{CLIP}^{aux}$	82.73	87.96	74.12	84.60
w/ $\mathcal{L}_{CE}^{clip} + \mathcal{L}_{CE}^{vis}$	81.22	87.49	71.48	82.98
w/ $\mathcal{L}_{CE}^{clip} + \mathcal{L}_{CE}^{vis} + \mathcal{L}_{dist}^{clip} + \mathcal{L}_{dist}^{vis}$	82.42	87.77	79.98	86.69
w/ all (a.k.a Lfl)	83.82	88.65	81.74	87.23

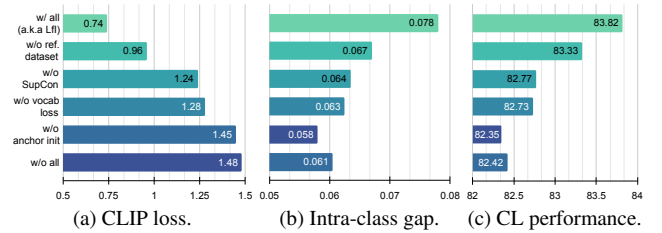


Figure 4. Token generation ablation on CIFAR-100. Intra-class gap is the similarity difference between in-class and unpaired samples.

sects our method’s learning objectives to understand their individual contributions. Using only the auxiliary CLIP loss with generated tokens ($\mathcal{L}_{CLIP}^{aux}$) yields limited improvement (75.83% on CIFAR-100), confirming that contrastive learning alone lacks class-level discrimination. However, combining it with classification loss ($\mathcal{L}_{CE}^{clip} + \mathcal{L}_{CLIP}^{aux}$) boosts performance to 82.73%, validating that generated tokens effectively bridge the distribution gap while classification provides necessary discriminative signals. Adding the vision classifier without distillation provides marginal gains via deep supervision, but mutual distillation dramatically improves fine-grained performance (71.48%→79.98% on CUB-200), demonstrating successful knowledge transfer between modalities. The complete Lfl framework achieves the best results, confirming the synergy between all components.

Token generation components. Fig. 4 analyzes the critical elements of our token generation process on CIFAR-100. Removing the class name anchor initialization has the most severe impact—without semantic grounding, optimization produces adversarial tokens that minimize CLIP loss (1.45) but destroy semantic coherence, leading to poor performance (82.35%). The reference dataset provides essential negative samples for contrastive learning; its removal increases CLIP loss from 0.743 to 0.960 and reduces the intra-class gap (the similarity difference between same-class samples with shuffled captions and random pairs) from 0.078 to 0.067, indicating degraded vision-language alignment while losing class-level consistency. Supervised contrastive and vocabulary alignment losses further refine token quality—their individual removal increases CLIP loss by ~0.5 and reduces

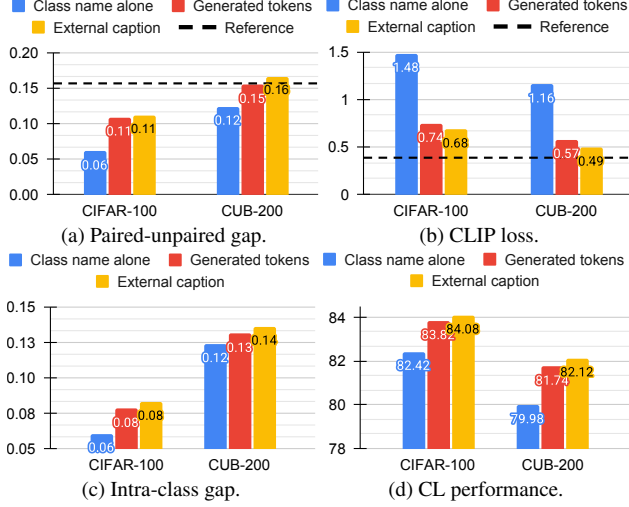


Figure 5. Generated tokens vs. external captions.

the intra-class gap, confirming that both components are essential for maintaining class-level consistency while preventing sample-specific overfitting.

Comparison with external captions. Fig. 5 validates that our self-generated tokens approach the quality of external captions from multimodal LLMs. On both datasets, generated tokens substantially reduce CLIP loss compared to class names alone (CIFAR-100: 1.48→0.74; CUB-200: 1.16→0.57), approaching external caption levels (0.68 and 0.49 respectively). The paired-unpaired similarity gap nearly doubles with our tokens (CIFAR-100: 0.06→0.11; CUB-200: 0.12→0.15), indicating restored vision-language alignment comparable to external captions. Crucially, the intra-class gap—measuring similarity between same-class samples with shuffled captions versus random pairs—also improves with our tokens (CIFAR-100: 0.06→0.08; CUB-200: 0.12→0.13), confirming that generated tokens maintain class-level coherence while adding instance-specific richness. This balanced improvement translates to performance: generated tokens achieve 83.82% on CIFAR-100 (vs. 84.08% with external captions) and 81.74% on CUB-200 (vs. 82.12%), demonstrating that CLIP can effectively mine its own knowledge for adaptation without requiring external models.

Mutual distillation strategies. Tab. 4 evaluates different distillation configurations to validate our adaptive bidirectional approach. Traditional symmetric mutual learning ($\mathcal{L}_{\text{mutual}}$ in Eq. (3)) between CLIP and vision heads provides negligible improvement, as it forces the stronger head to match weaker predictions. Unidirectional distillation from a frozen vision head to CLIP shows promise on CUB-200 (76.51%), transferring visual discrimination despite misalignment between the frozen classifier and evolving vision encoder. Allowing the vision head to train significantly improves transfer (80.82% on CUB-200), as both components remain aligned during adaptation. Our complete bidirectional approach

Table 4. Mutual learning strategies. Comparison of symmetric, unidirectional, and adaptive bidirectional distillation.

Loss	CIFAR-100		CUB-200	
	<i>last</i>	<i>avg</i>	<i>last</i>	<i>avg</i>
N/A (w/o mutual learning)	82.73	87.96	74.12	84.60
w/ $\mathcal{L}_{\text{mutual}}$	82.49	87.72	74.49	84.88
w/ $\mathcal{L}_{\text{CE}}^{\text{clip}} + \mathcal{L}_{\text{dist}}^{\text{clip}}$	82.86	88.05	76.51	85.54
w/ $\mathcal{L}_{\text{CE}}^{\text{clip}} + \mathcal{L}_{\text{CE}}^{\text{vis}} + \mathcal{L}_{\text{dist}}^{\text{clip}}$	83.29	88.28	80.82	86.96
w/ $\mathcal{L}_{\text{CE}}^{\text{clip}} + \mathcal{L}_{\text{CE}}^{\text{vis}} + \mathcal{L}_{\text{dist}}^{\text{clip}} + \mathcal{L}_{\text{dist}}^{\text{vis}}$	83.82	88.65	81.74	87.23

Table 5. Vision-only inference strategies. Performance using assembled task classifiers vs regenerated prototypes vs standard CLIP.

Method	CIFAR-100		CUB-200	
	<i>last</i>	<i>avg</i>	<i>last</i>	<i>avg</i>
Vision + task classifiers	81.78	87.58	80.19	86.59
Vision + SimpleCIL	82.40	87.69	80.95	87.02
Vision + Text (CLIP)	83.82	88.65	81.74	87.23

with adaptive weighting achieves optimal results (81.74% on CUB-200), enabling dynamic knowledge transfer between two heads based on instantaneous performance.

Vision-only inference analysis. Tab. 5 explores alternative inference strategies using only the vision encoder, testing whether mutual distillation creates equally capable branches. Direct assembly of all task-specific classifiers yields suboptimal results (81.78% on CIFAR-100), as classifiers trained on earlier tasks become misaligned with the continuously evolving vision encoder. Regenerating prototypes with SimpleCIL improves performance (82.40%) by ensuring classifier-encoder alignment, but still underperforms CLIP inference. The results confirm CLIP inference remains superior despite successful knowledge transfer through mutual distillation.

6. Conclusion

We presented Learning from Itself (LfI), enabling CLIP to mine its own internal knowledge for effective continual learning. By addressing the text distribution gap through self-generated pseudo-captions and leveraging complementary strengths of vision and text encoders via adaptive mutual distillation, our method achieves state-of-the-art performance without external assistance. Extensive experiments across diverse benchmarks validate that pretrained vision-language models contain rich knowledge that, when properly extracted, enables robust continual adaptation. Our work demonstrates that effective continual learning may lie not in external constraints or architectural modifications, but in better utilizing the knowledge already encoded within foundation models.

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